① ⓐ $(3+2+1)! = 6!$ ways to arrange
$= 720$

1 (if correct) no partial marking for this qn

ⓑ $3!\ 3!\ 2!\ 1! = 72$

order within math books

# of ways to order across categories

MCN
MNC
⋮ } $3!$

order within novels

1

ⓒ $3! \times 4!$

consider novels as an entity. then we can arrange across novels and the rest of the 3 books

arrangements within novels.

1

② $\frac{10!}{10} = 9!$

(1)

($10!$ arrangements in a row are possible since shifting the ordering gives same arrangement we need to divide by 10)

ⓐ E_i : event that ith couple sits next to each other

(2)

# of outcomes/arrangements such that the i-th couple stays together

$= 2 \times (9-1)!$

$= 2 \times 8!$.

$P(E_i) = \frac{2}{9}$.

(b) $P(E_1 \cap E_2 \cap \dots E_5) = \dfrac{2^5\, 4!}{9!}$.

(i)

□□□□□

think of each couple as an entity

# of arrangements in round table = 4!

arrangements within each entity = 2!

(c) $P(E_{i_1} \cap E_{i_2} \dots \cap E_{i_k}) = 2^k \dfrac{(9-k)!}{9!}$, $1 \le k \le 5$.

(i). Think of each of the k couples as one entity. So # of arrangements possible $= \dfrac{(10-k)!}{10-k} = (9-k)!$.

$$P(E_1 \cup E_2 \cup \dots E_5) = 5 \times \frac{2}{9} - \binom{5}{2} 2^2 \frac{7!}{9!}$$

(2)

$$+ \binom{5}{3} 2^3 \frac{6!}{9!} - \binom{5}{4} 2^4 \frac{5!}{9!}$$

$$+ \binom{5}{5} 2^5 \frac{4!}{9!}.$$

$$= \sum_{k=1}^{5} \binom{5}{k} (-1)^{k+1} 2^k \frac{(9-k)!}{9!}$$

(3) $P_X(k) = c$ for $c \in \{1, 2, \dots n\}$.

(a) $\sum_{k=1}^{n} P_X(k) = 1 \Rightarrow nc = 1 \Rightarrow c = \frac{1}{n}$.

(b) $E[X] = \sum_{k=1}^{n} k\, P_X(k) = \frac{1}{n} \sum_{k=1}^{n} k = \frac{n+1}{2}$.

$Var(X) = E[X^2] - (E[X])^2$

$E[X^2] = \sum_{k=1}^{n} k^2 P_X(k) = \frac{1}{\cancel{n}} \frac{\cancel{n}(n+1)(2n+1)}{6}$

$Var(X) = \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4} = \frac{(n+1)[(4n+2)-(3n+3)]}{12}$

$$\Rightarrow \quad Var(X) = \frac{n^2-1}{12} = \frac{(n+1)(n-1)}{2}.$$

4

a) $P(\{X=2\}) = F_X(2) - F_X(2^-)$

(1) $= 0.67 - 0.33 = 0.34.$

(1) b) $P(\{X<2\}) = 0.33.$

c) $P(\{X=2\} \cup \{0.5 \le X \le 1.5\})$

$= P(\{X=2\}) + P(\{0.5 \le X \le 1.5\})$

(1.5) $= 0.33 + F_X(1.5) - F_X(0.5) + \underbrace{P(\{0.5\})}_{=0}$

$= 0.33 + 0.33 - 8.33 \cdot 10^{-2}$ as $F_X(0.5) = F_X(0.5^-)$

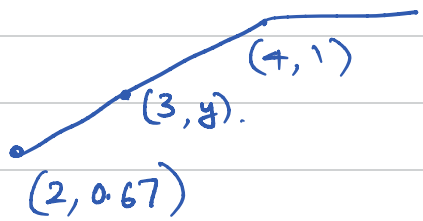
$= 0.67 - 0.0833$

d) $P(\{X=2\} \cup \{0.5 < X \le 3\}).$

$= P(\{0.5 < X \le 3\}).$

$= F_X(3) - F_X(0.5). = 0.83 - 0.0833$

(1.5) $= 0.75$



$(y - 0.67) = (1-y).$

$\Rightarrow \quad y = \frac{0.67+1}{2} = 0.83$